Aerofoil & Geometry Selection

1 Aerodynamics recap

Conservation of mass (mass is neither created nor destroyed):

$$\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2 = \text{constant} \xrightarrow{\rho = \text{const}} A_1 V_1 = A_2 V_2$$

Conservation of momentum: $F = \dfrac{d}{dt}(mV)$, which gives the Navier–Stokes equations.

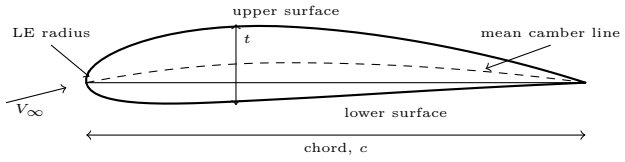
Common simplifying assumptions: steady flow; incompressible flow (ρ constant, valid for $M = V/a < 0.3$); inviscid flow ($\mu = 0$); no body forces.

- **Compressible** ($M > 0.3$): density varies, heat transfer matters, shock waves may form.
- **Viscous:** all real flows are viscous. Without viscosity there is no boundary layer, separation or turbulence (d'Alembert's paradox: zero drag). Only induced drag and supersonic wave drag come from inviscid theory.

$$p + \tfrac{1}{2}\rho V^2 = \text{constant} \qquad L' = \rho_\infty V_\infty \Gamma \qquad C_l = 2\pi\alpha$$

Bernoulli links velocity to pressure, and pressure to force. Stream function and velocity potential satisfy Laplace's equation ($\nabla^2\psi = 0$, $\nabla^2\phi = 0$), so elementary flows can be superimposed. Making the aerofoil surface a streamline gives the vortex sheet $\gamma \to \Gamma \to L'$ (Kutta–Joukowski). Thin aerofoil theory (steady, incompressible, inviscid) gives $C_l = 2\pi\alpha$.

2 Aerofoil geometry



- **Mean camber line:** joins the midpoints between the upper and lower surfaces.
- **Chord line:** straight line joining the end points of the mean camber line; its length is the chord c.
- **Thickness distribution:** distance between the surfaces, measured perpendicular to the camber line, $t = f(x)$. Thickness ratio t/c.
- **Leading edge:** defined by the leading-edge radius; sharp (or nearly sharp) for supersonic flight to avoid a bow shock.
- **Angle of attack** α: angle between the chord line and the oncoming flow V_∞.

An aerofoil affects cruise speed, take-off and landing distances, stall speed, handling near the stall, and aerodynamic efficiency in every phase of flight.

3 Pressure distribution and lift

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2}$$

- Angle of attack and camber make the air over the upper surface travel faster than below it: faster flow, lower pressure.
- $C_p = 1$ at the **stagnation point**. The suction peak marks the maximum velocity and minimum pressure.
- Ahead of the peak the pressure falls: **favourable gradient**, $dp/dx < 0$. Behind it the pressure rises: **adverse gradient**, $dp/dx > 0$, which drives separation.
- The area between the upper and lower C_p curves is the lift.

4 Force and moment coefficients

Lift is perpendicular to V_∞, drag is parallel to it; both, and the moment, are taken per unit span.

$$C_l = \frac{l}{\frac{1}{2}\rho_\infty V_\infty^2 c} \quad C_d = \frac{d}{\frac{1}{2}\rho_\infty V_\infty^2 c} \quad C_m = \frac{m}{\frac{1}{2}\rho_\infty V_\infty^2 c^2}$$

$$C_l, C_d, C_m = f(\alpha,\, Re,\, M_\infty)$$

$$Re = \frac{\rho_\infty V_\infty c}{\mu_\infty} = \frac{\text{inertia}}{\text{viscous}}, \qquad M_\infty = \frac{V_\infty}{a_\infty}$$

5 Characteristic curves

Lift curve (C_l vs α), linear part:

$$C_l = C_{l_\alpha}(\alpha - \alpha_{L=0}) = a\,(\alpha - \alpha_{L=0}), \qquad a_{\text{theory}} = 2\pi \text{ per rad}$$

A symmetric aerofoil has $\alpha_{L=0} = 0$. Lift peaks at $C_{l_{max}}$ at the stall angle.

Drag polar (C_l vs C_d): the tangent from the origin to the curve gives the maximum C_l/C_d, the measure of aerodynamic efficiency.

Pitching moment (C_m vs α): depends on the reference point. About the quarter chord, $C_{m_{0.25c}}$ is almost independent of α for most aerofoils at subsonic speed; the aerodynamic centre is generally at the quarter chord.

6 Reynolds and Mach number effects

Reynolds number decides whether the boundary layer is laminar or turbulent and whether it separates: laminar BL → transition → turbulent BL → separation and wake. A typical wing operates at $Re \approx 10^7$.

Mach number: the local flow over the curved surface is faster than the freestream. The **critical Mach number** M_{crit} is the freestream Mach number at which sonic flow first appears on the surface. Above it, a strong shock forms on the upper surface, separates the boundary layer, raises drag, lowers lift and changes the pitching moment.

7 NACA aerofoil families

NACA (forerunner of NASA) tested aerofoils systematically from the 1930s and grouped them into families.

Example	Meaning
4-digit	
NACA 4415	4: max camber 4% c 4: max camber at 40% c 15: thickness 15% c
NACA 0012	Symmetric (no camber), 12% thick. Common for tail surfaces.
5-digit	
NACA 23012	2: design $C_l = 2 \times 0.15 = 0.3$ 3: max camber at $3 \times 5 = 15\%$ c (moved forward for higher $C_{l_{max}}$) 0: standard camber line (1 = reflexed) 12: thickness 12% c
6-series	
NACA 63_2-212	6: six series (laminar flow) 3: minimum C_p at 30% c subscript 2: low drag for C_l within ± 0.2 of design 2: design $C_l = 0.2$ 12: thickness 12% c

The 6-series places maximum thickness far back to keep laminar flow and lower drag; better at high speed than the 4- and 5-digit families.

8 Aerofoil design and the design lift coefficient

Selection criteria: cruise drag, stall and pitching-moment behaviour, thickness for structure and fuel, ease of manufacture. Today aerofoils are often custom designed (potential flow outside the boundary layer, viscous corrections inside), for a design point such as maximum lift, maximum thickness, transonic flight, laminar flow, low Re or low pitching moment.

The **design lift coefficient** is the C_l at which the aerofoil has its best L/D (tangent from the origin). The aircraft should fly its mission at or near it.

$$W = L = qSC_L \approx qSC_l \quad \Rightarrow \quad C_l = \frac{1}{q}\frac{W}{S}$$

As fuel burns, W/S falls, so q must fall and the aircraft climbs: the *cruise climb* for maximum range. In practice the design C_l is about **0.5**, often taken from experience or a successful design.

9 Stall

Type	t/c	Mechanism	Lift / moment
Gradual (TE)	$> 14\%$	Separation starts at the TE (around 10°) and moves forward; rounded LE	Gradual lift loss, small C_m change; forgiving for trainees
Abrupt (LE)	6–14%	Separation near the nose reattaches, then suddenly fails to; sharper LE	Abrupt loss of lift and change in C_m
Thin aerofoil	$< 6\%$	LE bubble stretches to the TE; $C_{l_{max}}$ when it reaches it	Smooth lift loss, large C_m changes

Section stall and $C_{l_{max}}$ depend on thickness ratio, camber line shape and leading-edge shape and radius. The flat plate stalls earliest and worst.

10 Effects of thickness, camber and leading edge

Thickness ratio affects $C_{l_{max}}$, stall, drag and structural weight:

- $t/c \uparrow$: drag $\uparrow$ (more separation); $M_{crit} \downarrow$.
- Wing weight $\propto 1/\sqrt{t/c}$: halving t/c adds about **41%** to wing weight. The wing is about 15% of empty weight, so empty weight rises by about 6%, which the sizing equation amplifies.

Camber sets the lift at zero angle of attack. Adding camber always raises $C_{l_{max}}$, more so for thin sections than thick ones.

Leading edge: measured by z_5/t (thickness at 5% chord over maximum thickness). A large leading-edge radius gives a high $C_{l_{max}}$ at low speed.

11 Special-purpose aerofoils

- **Supercritical:** spreads lift along the chord to lower upper-surface velocity, giving a weak shock, less separation and a higher M_{crit}. Less camber, larger LE radius, flat upper surface, concave rear lower surface. Used by transonic airliners.
- **Laminar:** pressure falls continuously towards the TE, keeping laminar flow over 50–70% of the wing for low skin friction. Drawbacks: lower $C_{l_{max}}$ and high sensitivity to surface finish.
- **Rotorcraft:** high α near stall, transonic tips, high lift with low drag and low (often reflexed) pitching moment.

Thick, cambered, round-nosed sections suit low speed; thin, sharp-nosed sections suit high and supersonic speed.

12 High-lift devices

Device	Effect
Plain flap	Hinged rear of the section; increases effective camber. Up to about 15° without separation.
Split flap	Lower surface only; larger $\Delta C_{l_{max}}$ than plain, but much higher drag (large wake).
Slotted flap	Slot feeds high-energy air to the upper surface (BL control); much higher $\Delta C_{l_{max}}$, lower drag; large negative C_m.
Fowler flap	Slotted flap that also moves aft, increasing wing area; highest ΔC_l.
LE slat	Small aerofoil ahead of the LE, fixed or retractable; delays LE stall, $\Delta C_{l_{max}}$ up to about 0.5.
Slots	Similar to a slat; BL control, but raise minimum drag at low α.
BL control	Blowing or suction; raises $C_{l_{max}}$, suction can also delay transition.

Flap deflection: makes $\alpha_{L=0}$ more negative with little change in lift-curve slope, raises $C_{l_{max}}$, lowers the stall angle, raises C_d and lowers L/D. Lift gains come at the cost of drag and weight; flaps and slats retract in cruise.